MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, GERMANIUM, HIGH-POWER TYPE 2N665

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detailed requirements for a high-power, germanium, PNP, Transistor.

1.2 Physical dimensions. See figure 1 (TO-3).

1.3 Maximum ratings.

P_T 1/ T_C = 25° C	V_{CBO}	V_{EBO}	V_{CEO}	I_E	I_B	T_{stg}
W	Vdc	Vdc	Vdc	Adc	Adc	° C
35	-80	-40	-40	5	1	-65 to +100

1/ Derate linearly 0.466 W/°C for $T_C > 25°$ C.

1.4 Primary electrical characteristics.

	h_{FE}	h_{FE}	f_{hfe}	$V_{CE}(sat)$	V_{BE}
	V_{CE} = -2 Vdc I_C = -0.5 Adc	V_{CE} = -2 Vdc I_C = -2 Adc	V_{CE} = -14 Vdc I_C = -2 Adc	I_B = -220 mAdc I_C = -3 Adc	V_{CE} = -2 Vdc I_C = -2 Adc
			kc	Vdc	Vdc
Min	40	20	20	---	---
Max	80	--	--	-0.9	-1.5

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for

FSC 5961

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STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500 and as follows:

V_{CEO} - - - - - - - Collector to emitter voltage (static), base open.

3.3 Design and construction. Transistors shall be of the design, construction, and physical dimensions shown on figure 1.

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I, II, and III.

3.5 Marking. The following marking specified in MIL-S-19500 may be omitted from the body of the transistor at the option of the manufacturer:

(a) Country of origin.
(b) Manufacturer's identification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I, II, and III.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of groups A, B, and C inspections.

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

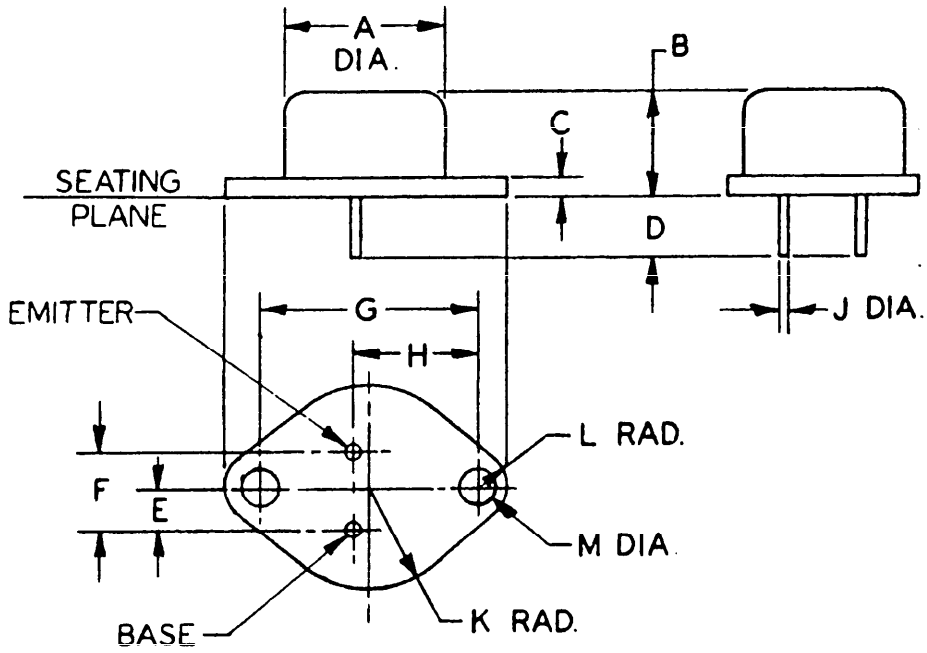
4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

4.3.3 Group C inspection. Group C inspection shall concist of the examinations and tests specified in table III. This inspection shall be conducted on the initial lot and thereafter every 6 months during production.

4.4 Methods of examination and test. Methods of examination and test shall be as specified in tables I, II, and III.

4.5 Inspection lot. Inspection lot shall be as defined in MIL-S-19500 except that lot accumulation period requirements shall be six months in lieu of six weeks.

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LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
A	---	.875	---	22.23	
B	.250	.450	6.35	11.43	
C	---	.135	---	3.43	
D	.312	---	7.92	---	3
E	.205	.225	5.21	5.72	
F	.420	.440	10.67	11.18	
G	1.177	1.197	29.90	30.40	
H	.655	.675	16.64	17.15	2
J	.038	.043	.97	1.09	3
K	---	.525	---	13.34	
L	---	.188	---	4.78	
M	.151	.161	3.84	4.09	

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. This dimension should be measured at points .050 (1.27 mm) to .055 (1.40 mm) below seating plane. When gage is not used, measurement will be made at seating plane.
3. Two leads.
4. Collector shall be electrically connected to the case.

FIGURE 1. Physical dimensions of transistor type 2N665 (TO-3).

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TABLE I. Group A inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1			10				
Visual and mechanical examination	2071			---	---	---	---
Subgroup 2			5				
Emitter to base cutoff current	3061	Bias cond. D; V_{EB} = -40 Vdc		I_{EBO}	---	-2	mAdc
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -2 Vdc		I_{CBO}	---	-50	μAdc
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -60 Vdc		I_{CBO}	---	-2	mAdc
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -80 Vdc		I_{CBO}	---	-10	mAdc
Forward-current transfer ratio	3076	V_{CE} = -2 Vdc; I_C = -0.5 Adc		h_{FE}	40	80	---
Forward-current transfer ratio	3076	V_{CE} = -2 Vdc; I_C = -2 Adc		h_{FE}	20	---	---
Base emitter voltage (nonsaturated)	3066	Test cond. A; V_{CE} = -2 Vdc; I_C = -2 Adc		V_{BE}	---	-1.5	Vdc
Collector to emitter voltage (saturated)	3071	I_C = -3 Adc; I_B = -220 mAdc		$V_{CE(sat)}$	---	-0.9	Vdc
Floating potential	3020	V_{CB} = -80 Vdc; voltmeter input resistance = 10 megohms min		V_{EBF}	---	-1	Vdc
Breakdown voltage, collector to emitter	3011	Bias cond. D; I_C = -300 mAdc		BV_{CEO}	-40	---	Vdc
Subgroup 3			5				
Small-signal short-circuit forward-current transfer-ratio cutoff frequency	3301	V_{CE} = -14 Vdc I_C = -2 Adc		f_{hfe}	20	---	kc
High-temperature operation: Emitter to base cutoff current	3061	T_C = 71° C; Bias cond. D; V_{EB} = -30 Vdc		I_{EBO}	---	-2	mAdc
Collector to base cutoff current	3036	T_C = 71° C; Bias cond. D; V_{CB} = -30 Vdc		I_{CBO}	---	-2	mAdc
Low-temperature operation: Forward-current transfer	3076	T_C = -55° C; V_{CE} = -2 Vdc; I_C = -0.5 Adc		h_{FE}	30	---	---

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TABLE II. Group B inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1			20				
Physical dimensions	2066	(See figure 1)		---	---	---	---
Subgroup 2			15				
Solderability	2026	Dwell time = 10 ±1 sec; omit aging		---	---	---	---
Thermal shock (temperature cycling)	1051	Test cond. B except high temp. = + 100 $^{+3^\circ}_{-0}$ C		---	---	---	---
Thermal shock (glass strain)	1056	Test cond. B		---	---	---	---
Terminal strength (tension)	2036	Test cond. A; weight = 10 lbs; time = 15 ±3 sec each lead		---	---	---	---
Terminal strength (lead torque)	2036	Test cond. D_1; torque = 6 in-oz; time = 15 ±3 sec each lead		---	---	---	---
Seal (leak-rate)	---	Method 112 of MIL-STD-202, test cond. C, procedure III; test cond. B for gross leaks		---	---	$5x10^{-7}$	atm cc/sec
Moisture resistance	1021	Omit initial conditioning		---	---	---	---
End points: Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -60 Vdc		I_{CBO}	---	-2	mAdc
Forward-current transfer ratio	3076	V_{CE} = -2 Vdc; I_C = -0.5 Adc		h_{FE}	40	80	---
Subgroup 3			15				
Shock	2016	Nonoperating; 500 G; for 1 msec; 5 blows in each orientation: X_1, Y_1, Y_2, and Z_1		---	---	---	---
Vibration fatigue	2046	Nonoperating; 10 G		---	---	---	---
Vibration, variable frequency	2056	Nonoperating; 10 G		---	---	---	---
Constant acceleration	2006	10,000 G; in each orientation: X_1, Y_1, Y_2, and Z_1		---	---	---	---
End points: (Same as for subgroup 2)							

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TABLE II. Group B inspection - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 4			15				
Salt atmosphere (corrosion)	1041			---	---	---	---
End point: (same as for subgroup 2)							
Subgroup 5			λ = 15				
High-temperature life (nonoperating)	1031	T_{stg} = +100° C		---	---	---	---
End points: Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -60 Vdc		I_{CBO}	---	-4	mAdc
Forward-current transfer ratio	3076	V_{CE} = -2 Vdc; I_C = -0.5 Adc		h_{FE}	32	96	---
Subgroup 6			λ = 10				
Steady state operation life	1026	T_C = + 75° C; P_T = 10 W		---	---	---	---
End points: (Same as for subgroup 5)							

TABLE III. Group C inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1			15				
Barometric pressure, reduced (altitude operation)	1001	Normal mounting; pressure = 8 mm Hg for 60 sec. min.		---	---	---	---
Measurement during test:							
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -80 Vdc		I_{CBO}	---	-10	mAdc
Thermal resistance	3151			θ_{J-C}	---	2	°C/W

5. PREPARATION FOR DELIVERY

5.1 See MIL-S-19500, section 5.

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6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - SH
Air Force - 11

Review activities:
Army - EL, MU, MI,
Navy - SH
Air Force - 11, 17, 85

User activities:
Army - EL, SM
Navy - CG , MC, WP
Air Force - 14, 19

Preparing activity:
Army - EL

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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